

DEPARTMENT OF COMPUTER ENGINEERING



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CHEMBUR, MUMBAI – 400088.

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Project Report on

**Roller Coaster
using Blender**

by

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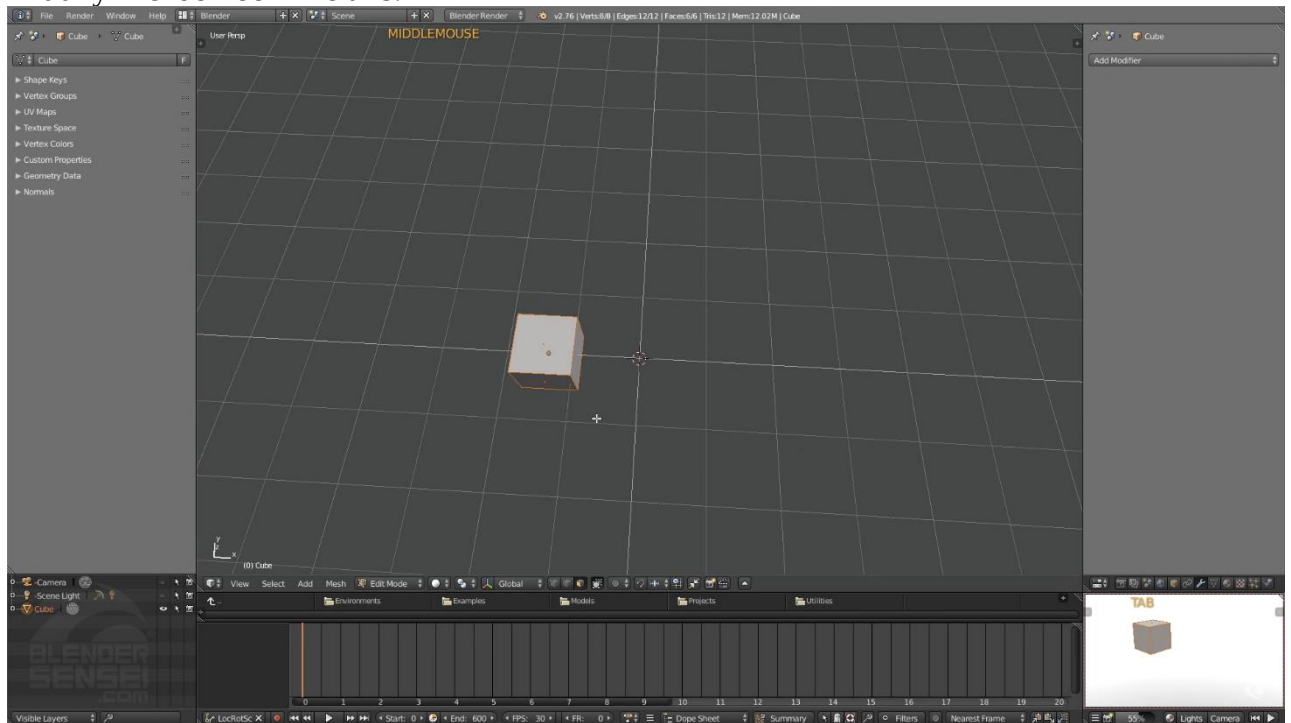
THEORY

I. Introduction.

What's Blender?

- Blender is the free and open source 3D creation suite. It supports the entirety of the 3D pipeline—modeling, rigging, animation, simulation, rendering, compositing and motion tracking, even video editing and game creation.
- Blender is cross-platform and runs equally well on Linux, Windows, and Macintosh computers. Its interface uses OpenGL to provide a consistent experience. To confirm specific compatibility, the list of supported platforms indicates those regularly tested by the development team.
- As a community-driven project under the GNU (General Public Licence), the public is empowered to make small and large changes to the code base, which leads to new features, responsive bug fixes, and better usability. Blender has no price tag, but you can invest, participate and help to advance a powerful collaborative tool: Blender is your own 3D software.

Initially Blender look like this:-



II. Tools Used

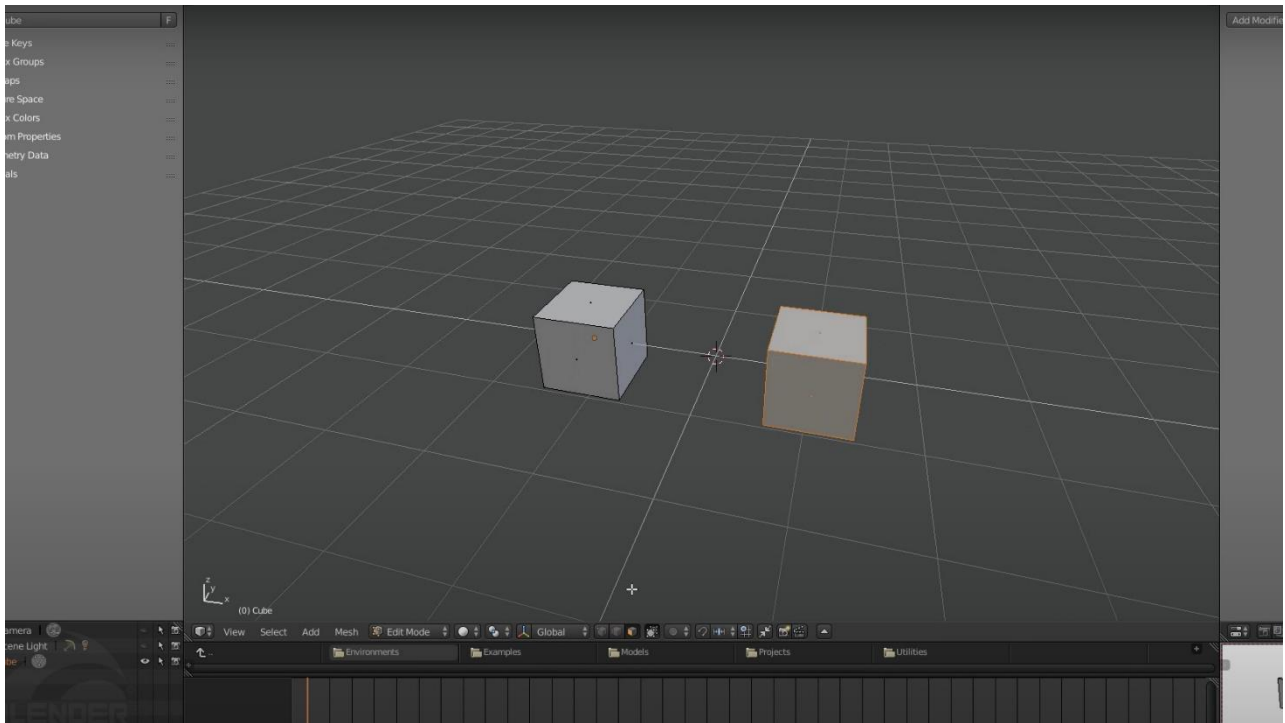
In this project we have used Blender Software.

III.Steps

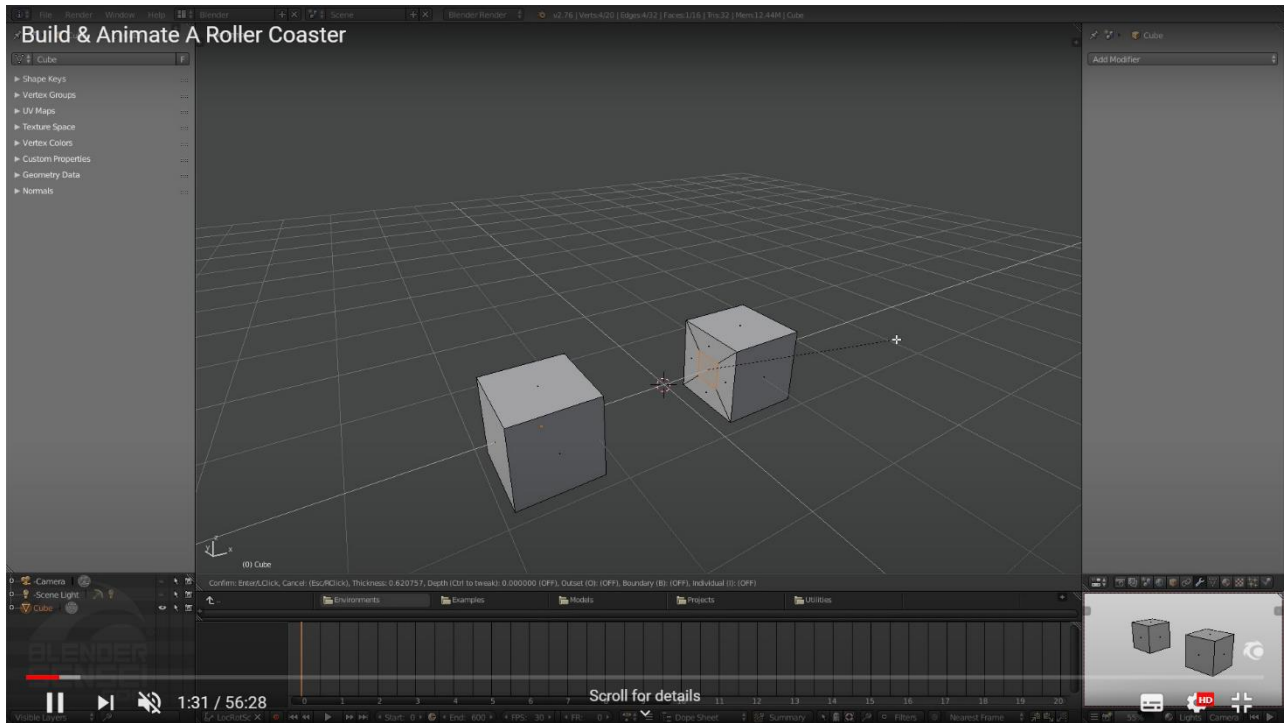
Under Following steps we have created our project

Steps to create a normal tracks:-

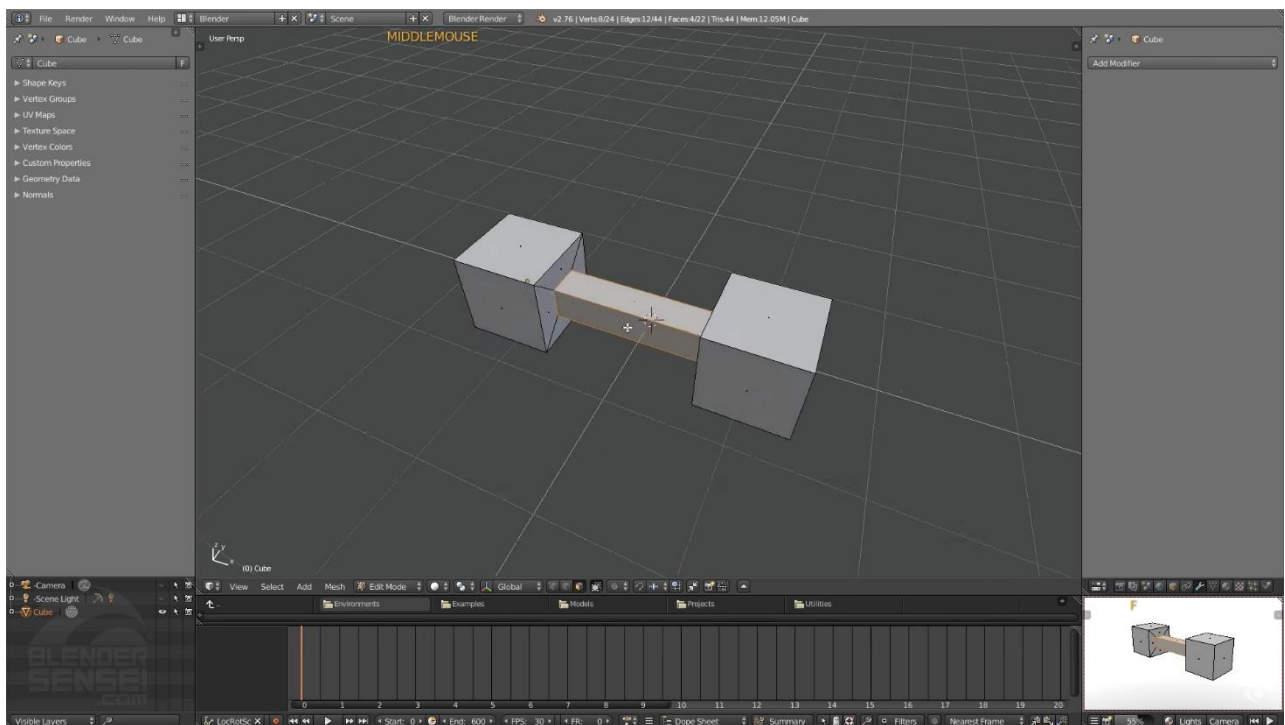
1. Press S to scale , G to grab middle mouse to constrain.
2. Press D to duplicate cube.



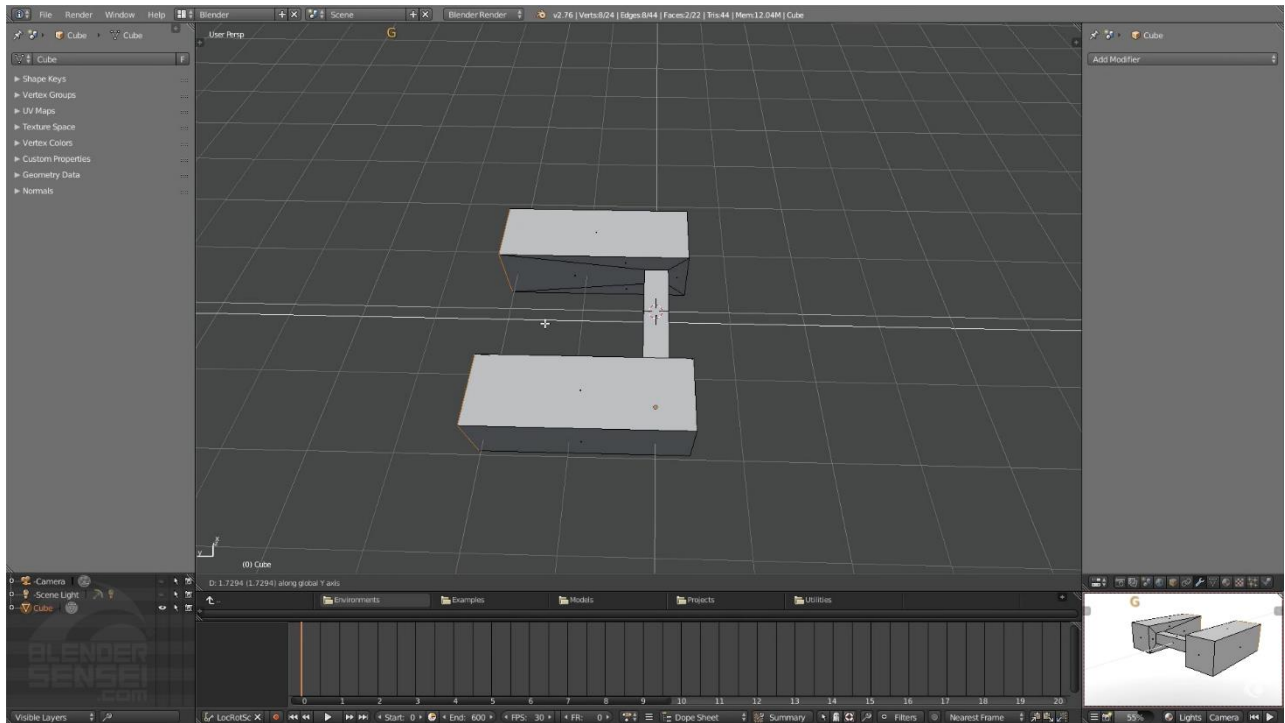
3. Select inward faces of cubes facing towards each other.
4. Press shift + E and inset the arrow and bring it downwards.



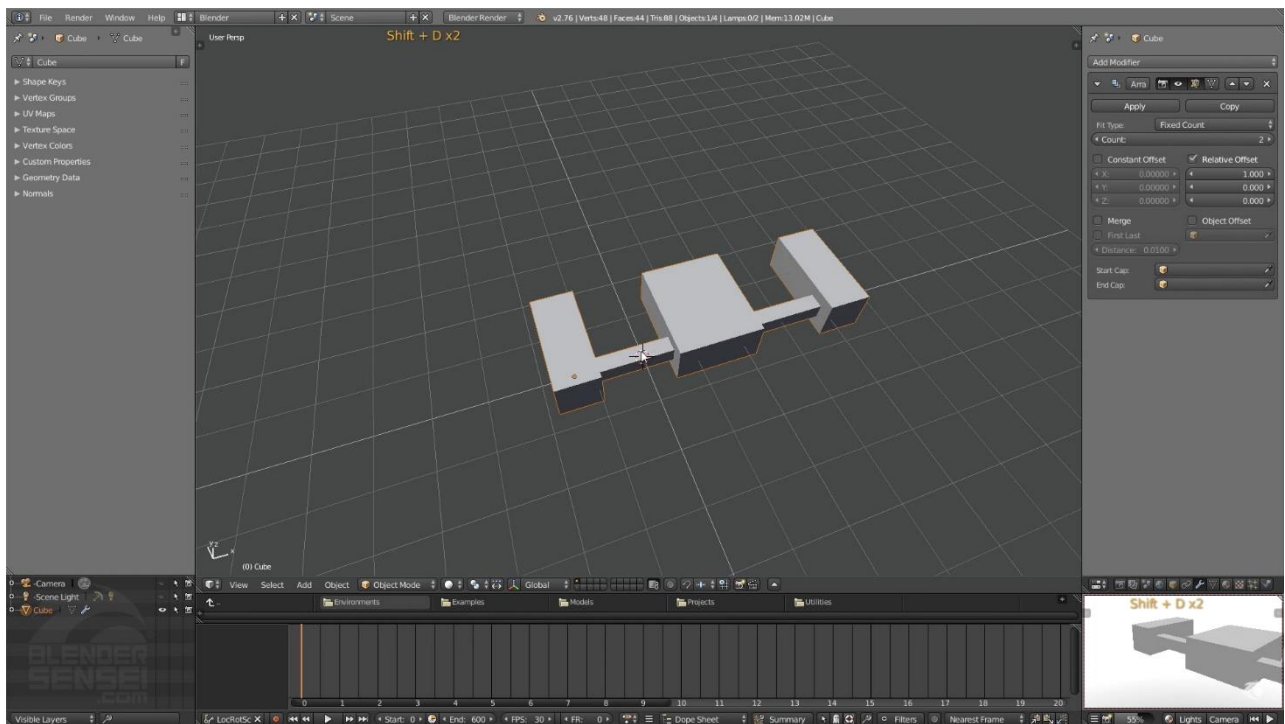
5. Press Shift+R to repeat the last tool used.
6. Shift+select other face, process F to bridge both cube.



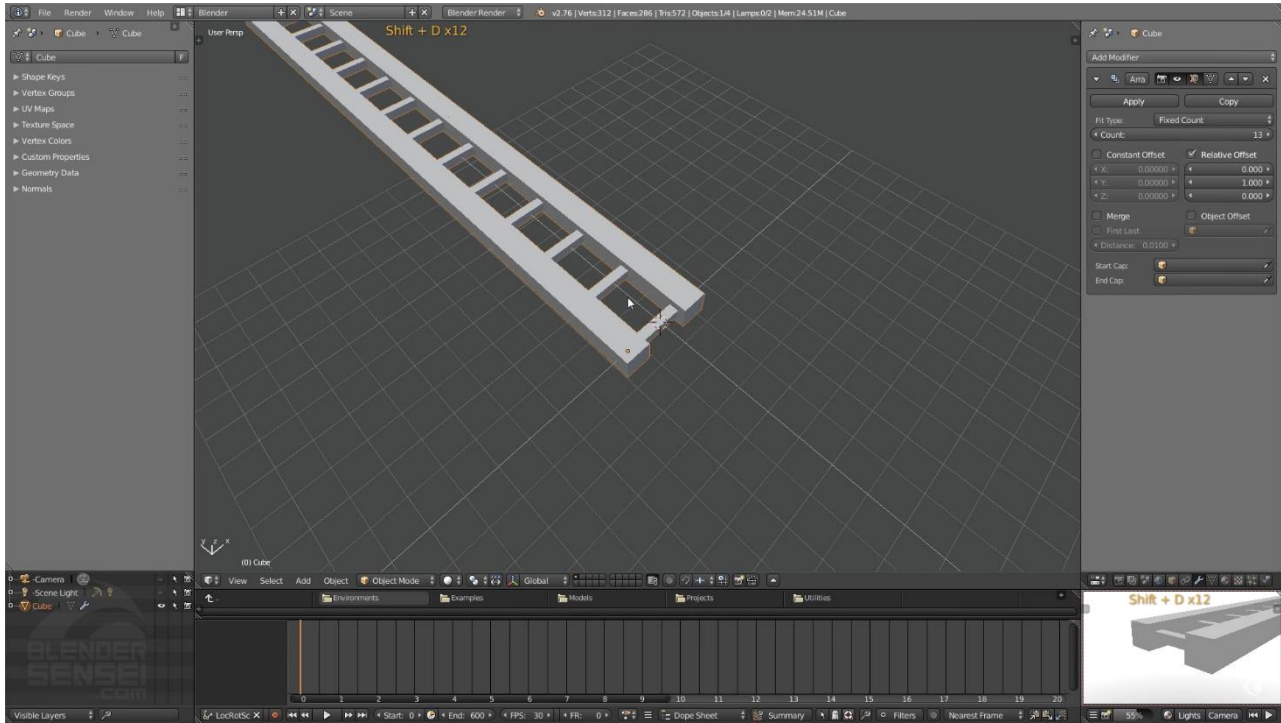
7. Then Shift + Select both cubes faces and press G to grab and make it longer.



8. Shift + D to add like same as before.



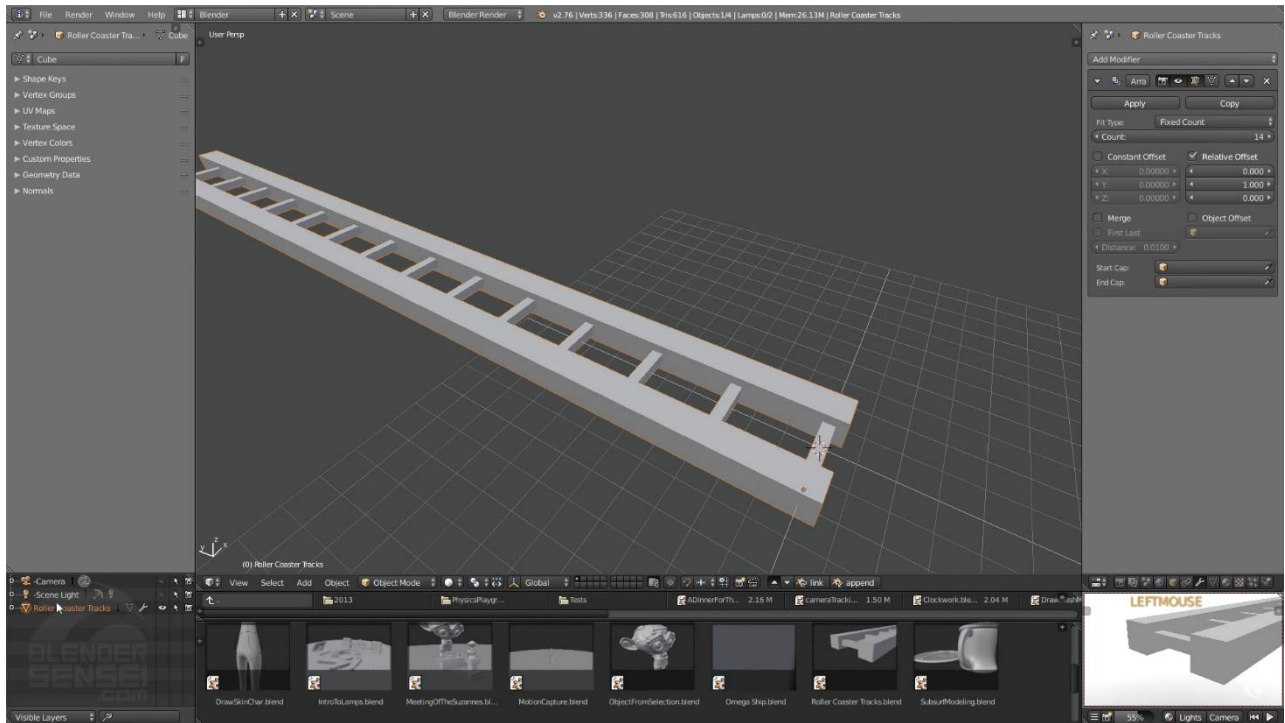
9. Shift + E and Shift + D for how much we want to extend the track.



10. Shift + E to keep adding them.

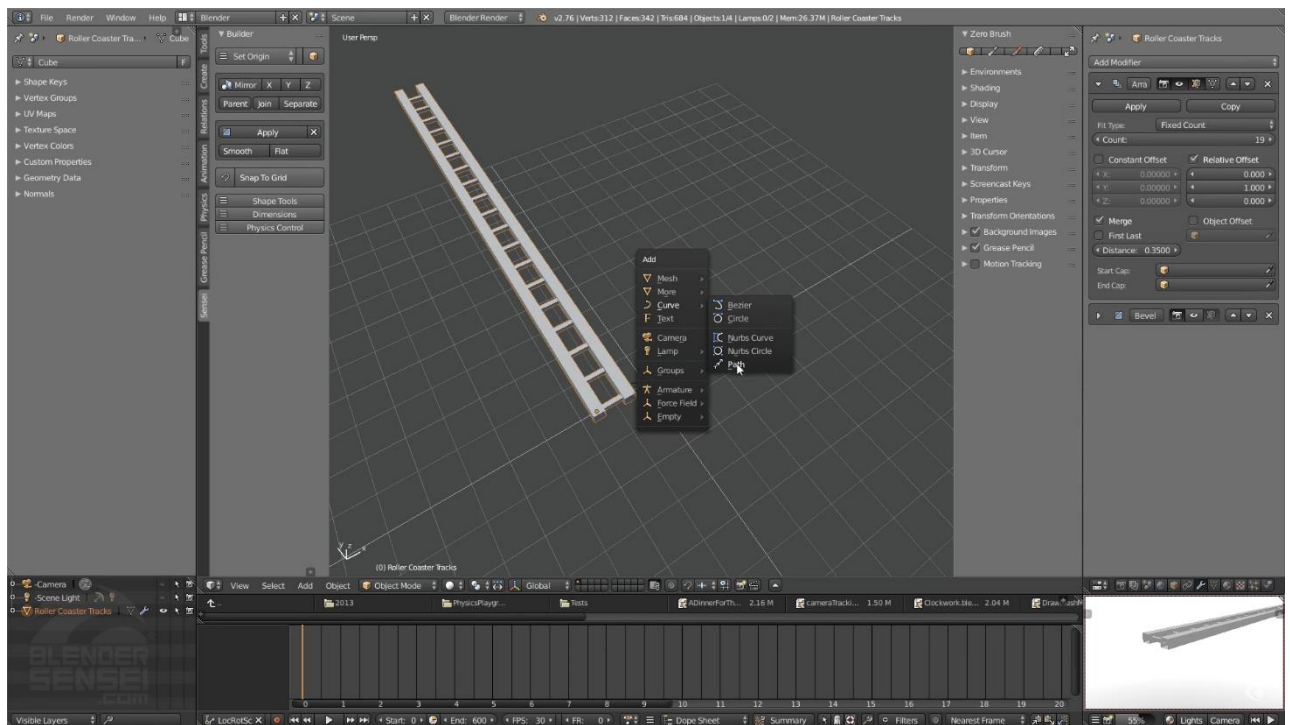
11. Alt+Shift+E to change their 3D direction.

12. On the top right corner of the box there is count of track option we can also increase or decrease count of tracks from it.

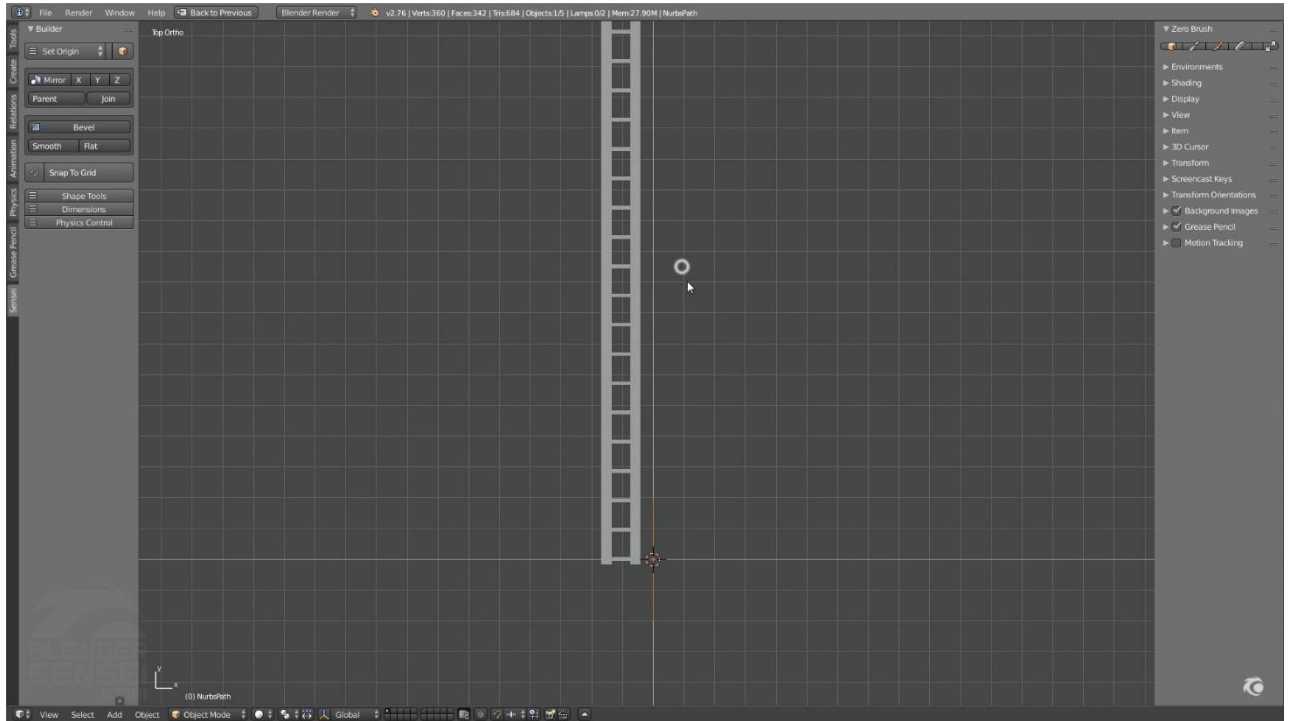


Steps to give path for tracks:-

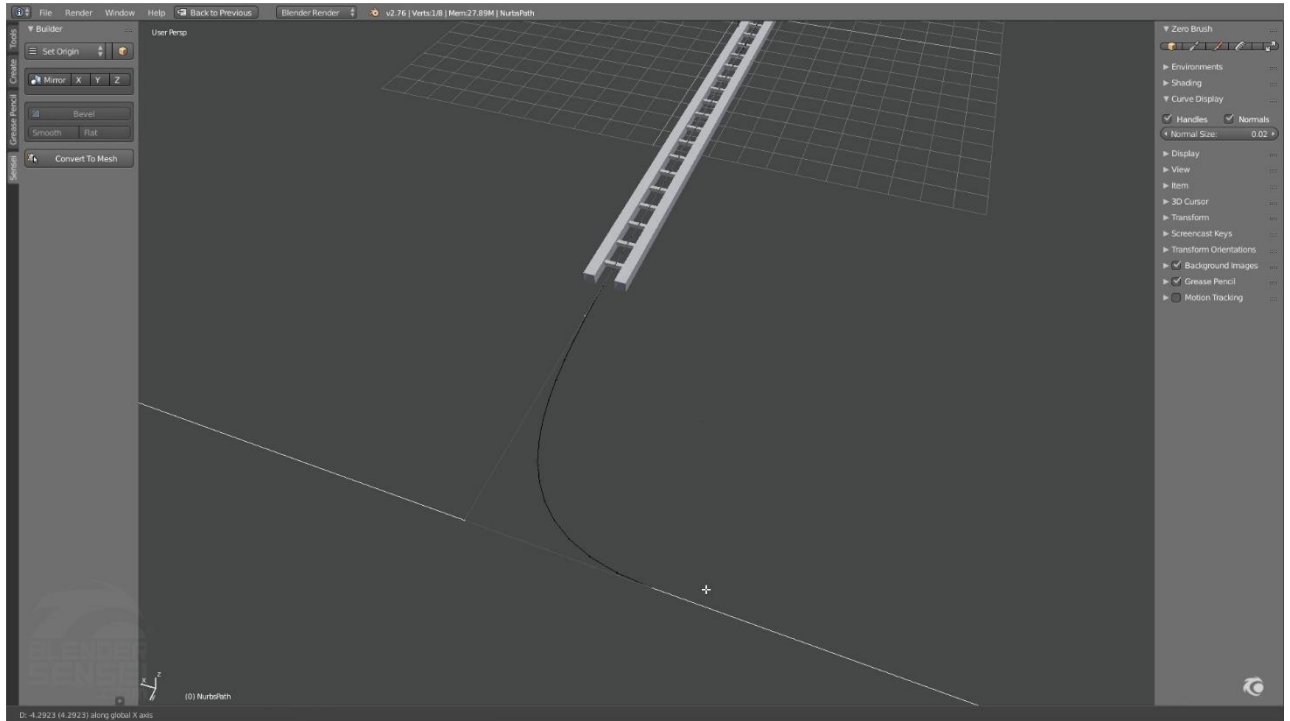
1. Shift + a and under curve option select Bezier or circle curve or NURBS.



2. Press V to go into top view.



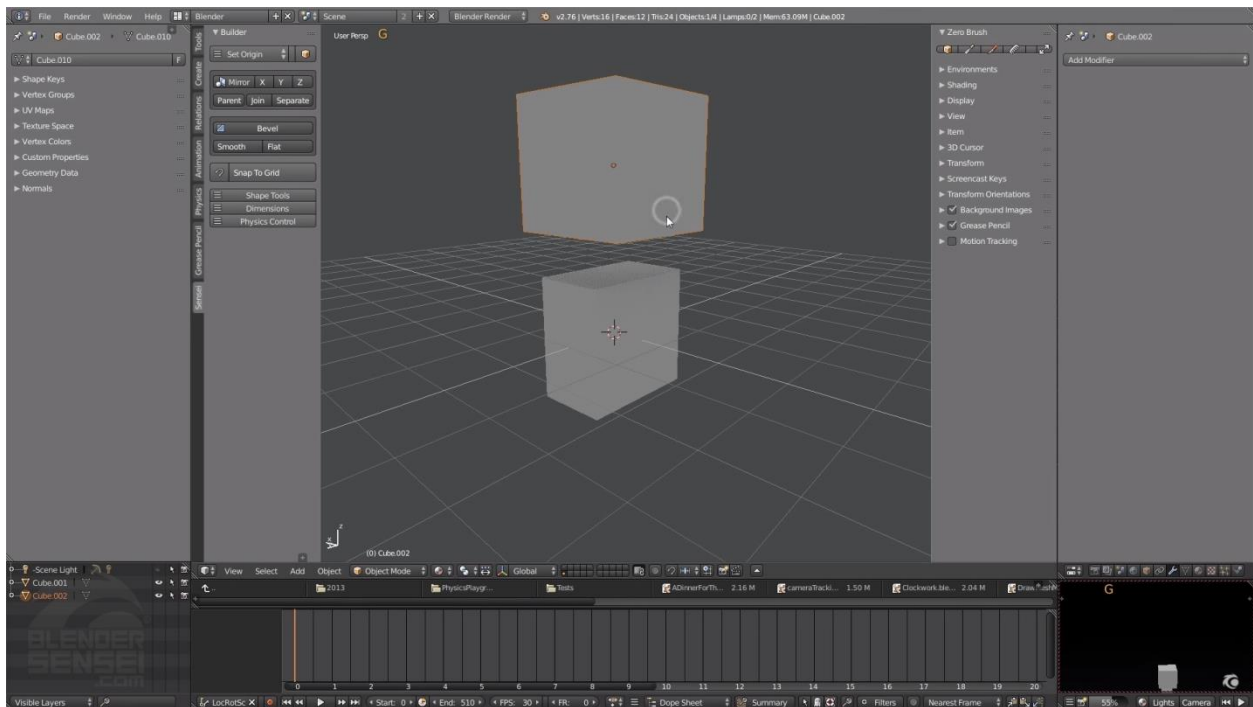
3. Press R to rotate and hold control.
4. Press G to grab them onto tracks.
5. To curve the path or to bend the path just select last point of line and grab according to how much we want.



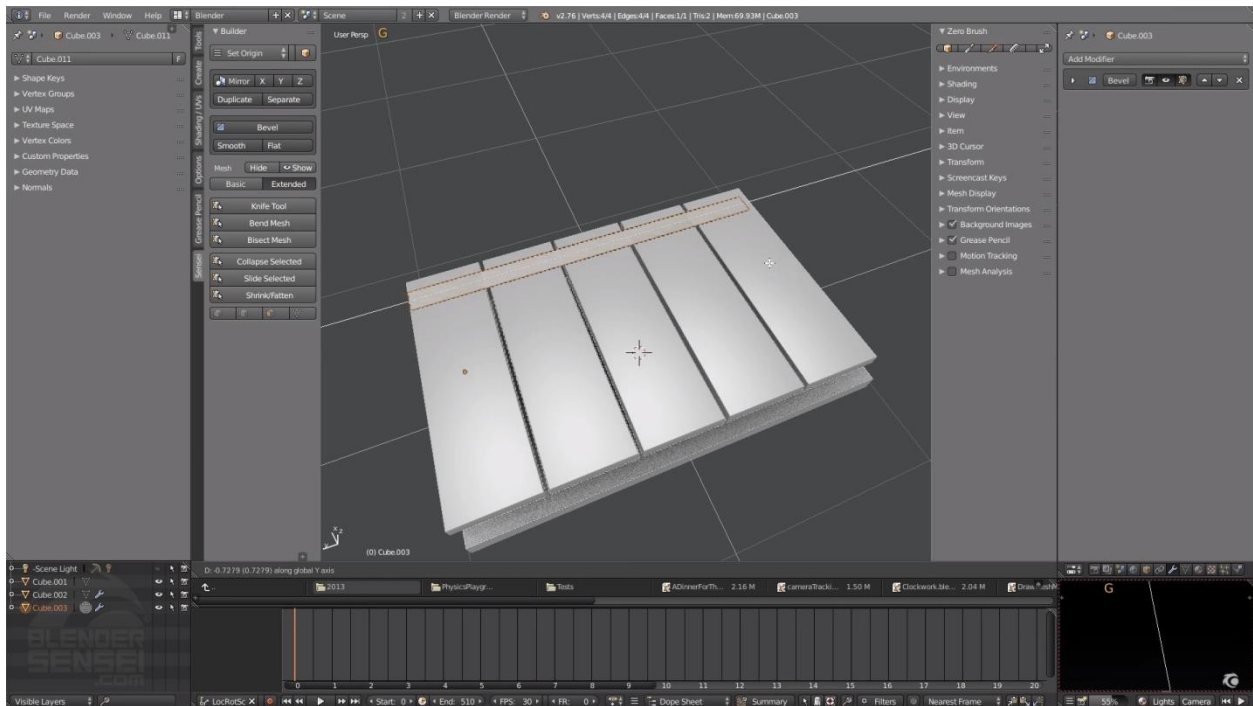
Steps for Rail Cart Addition:-

1. We select the track and press alt+a to add mesh here we add a cube mesh, then we move the cube to a separate layer by selecting object+m so that we a clear object without any

interruption. This object is a reference base of track.

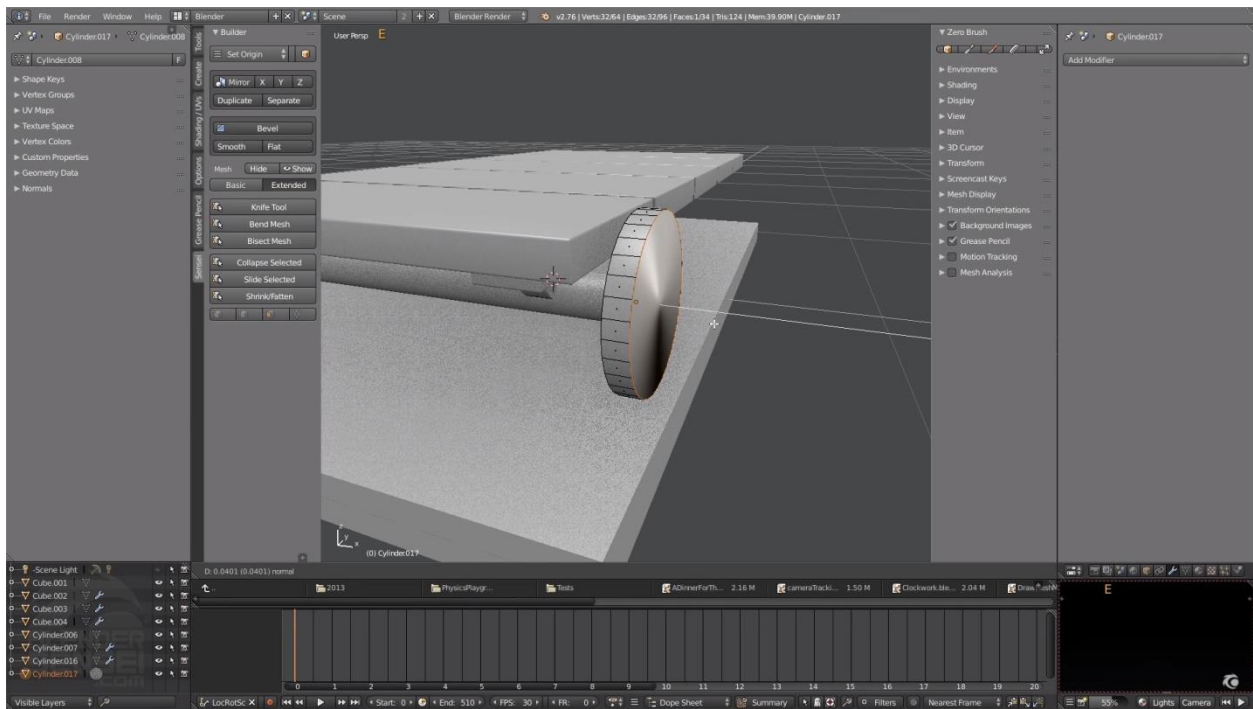


2. We thin scaled the cube by select+s and duplicate the cube by select+d into 5 same objects and attach it with a strip.



3. Now for the wheels we create a cylinder (alt+a) and kind of attach to the base. Now we duplicate the cyclinder and overlap it with each other and scale to make a latch and extrude

upwards and scale.

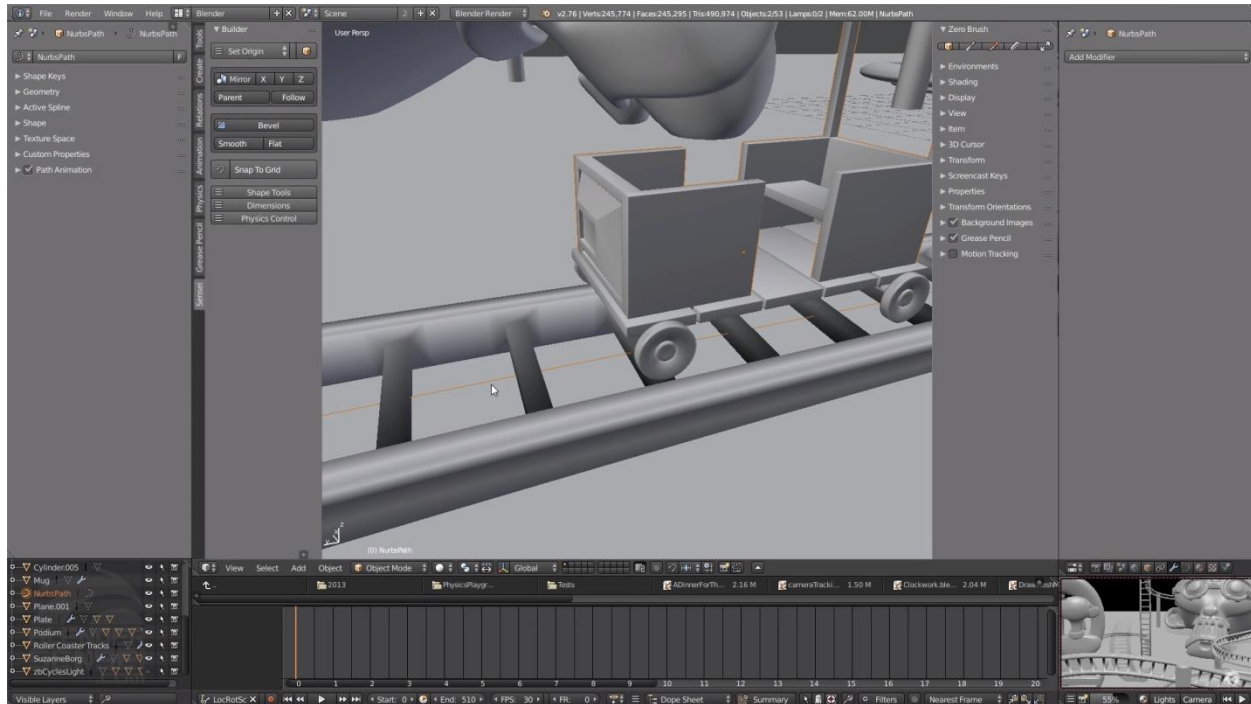


4. For wheels we select the in-phase, extend outwards and scale and duplicate (select+d) to other side as well .
5. The cart is ready and for upper design we added same cubes for body of cart.



Steps for giving motion to the cart:-

1. Grab the cart from the separately created layer nselect+g and align it over the track.
2. Now select the cart and shift+select the path and press follow from builder option it will pop up as you shift select.



Steps for making the camera follow the train:-

1. We actually making the camera follow the path.
2. Like the train we shift+select the path and under camera setting press follow path.



Conclusion

The main aim of our project was to study the basics of blender software, its functions and methods. We studied the following topics to implement our project.

- How to use the array modifier.
- Array modifier hotkeys.
- Loading and saving to the assets library
- Merging objects within an array.
- Using the curve modifier with an array.
- Using Edge loops to smooth curve modifier objects .
- Importing assets from your library into your scene.
- Deleting path curve points.
- Making a curve or path auto loop into itself.
- Building a cart for a roller coaster.
- Create objects from your mesh selection.
- Make an object follow a path.
- Manually adjusting path objects.
- Make multiple objects follow path.
- Camera follow path and track object.

We took help of sensei format also. Sensei Format comes with a free Blender **asset** library system which allows you to maximize reusability of your files, textures, images, models, imports and more. The Blender software has plethora of features like bridging objects, curving, smoothening them to create the required simulation. It also has features for following a specific object in motion/simulation and viewing it in a specific perspective. With the help of these effects and features on regular shapes, we were able to create objects and surroundings for the simulation of the project.

We also learned about the organization of the Blender Foundation and found this organization especially interesting. Most of the staff biographies provide that the Blender staff is made up of many young adults. The work of other young adults of my generation is increasingly relevant, and we are fascinated to learn that these people have created a program so intricate and detailed by working together so quickly after entering the work field.

